

LING185A: Computational Linguistics 1

Fall 2024

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Lectures: Royce 154
Mon/Weds 2-3:50pm

Sections: 1A: Thurs 2-2:50pm, Public Affairs 2317
1B: Thurs 3-3:50pm, Public Affairs 1264

Course Description

Computational linguistics is a large, multi-faceted, and rapidly expanding field. There are many different kinds of work that might be called “computational linguistics,” which vary in their goals (e.g. building a useful gadget or investigating a cognitive phenomenon) and linguistic domain (e.g. sounds, words, sentences). However, there are certain core analytical concepts, tools, and techniques that frequently appear throughout these different approaches. The relevant foundational concepts concern the computational properties of the kinds of systems that are studied by linguists: namely, grammars. This course aims to highlight these common ideas, starting with their simplest instantiations and gradually building up towards the more complex cases. The simplest cases will parallel certain things you may have learned about in mathematics or computer science courses, and the more complex cases will look similar to things you have learned about in linguistics courses. An important goal is to highlight the connections between these areas.

Some of the biggest ideas that will come up repeatedly in this course include: recursive generation of infinitely many expressions by a finite machine; interchangeability/intersubstitutability of subexpressions within larger expressions; and the relationship between discrete structures and probabilistic models.

Prerequisites: LING120B and either CS32 or Program in Computing 10C

Course Goals

- To gain familiarity with foundational concepts in formal language theory, particularly concerning the properties and limitations of finite-state and context-free grammars
- To develop an understanding of the relationship between grammatical structure, recursion, and dynamic programming
- To develop fluency in understanding formal mathematical descriptions of grammatical systems, and in writing programs to implement simple models of these systems
- To gain an appreciation of the insights that can be gained by studying the human mind through the lens of formal/computational models

Course Requirements

Homework assignments	70%
Final project	30%
Total	100%

Course Schedule

This schedule is tentative and may be adjusted as the quarter progresses.

<i>Week</i>	<i>Topic</i>	<i>Assignments</i>
<i>Setting the Scene: Core Computational Concepts</i>		
1	Haskell basics, variables and substitution	
2	Infinity and recursion	HW 1 due
<i>Finite-State Automata (FSAs)</i>		
3	Introduction to FSAs	HW 2 due
4	More on FSAs: determinism, closure properties, regular expressions	HW 3 due
5	Variants of FSAs: strict locality, weighted, transducers, semirings	HW 4 due
<i>Context-Free Grammars (CFGs) and beyond</i>		
6	Introduction to CFGs	HW 5 due
7	Parsing with CFGs (<i>no class Mon. Nov 11</i>)	HW 6 due
<i>Beyond Context-Free</i>		
8	Parsing with CFGs continued, intro to tree grammars	HW 7 due
9	More on tree grammars (<i>no class Weds. Nov 27</i>)	
<i>Probabilistic Grammars and Learning</i>		
10	Intro to probabilistic FSAs and CFGs, probabilistic learning	HW 8 due
<i>Finals Week: Final projects due Dec 13th</i>		

Course Materials and Resources

Readings

There is no required textbook for this course. All materials (readings, assignments, lecture and section handouts, and code snippets) will be distributed through BruinLearn. Two well-known textbooks that are often used for introductory computational linguistics courses are Jurafsky & Martin's *Speech and Language Processing* (Prentice Hall) and Manning & Schütze's *Foundations of Statistical Natural Language Processing* (MIT Press). Both take a more application-oriented approach than we'll be taking in this course, but you can find relevant supplementary material in both of them if you wish.

Software

Many of the homework exercises will involve writing or modifying small programs. The programming language we will use is Haskell. You will need access to a computer with a text editor and Haskell installed (<https://www.haskell.org/downloads/>).

Note that our use of Haskell in this course is a means to an end. It's a tool for sharpening our understanding of certain concepts, and those concepts, rather than Haskell itself, are the real focus of the course. Because of the secondary role of Haskell itself, I do not recommend consulting programming textbooks or tutorials that aim to teach you the language, at least not until after the first few weeks. My aim is to provide enough information in lectures and sections for this not to be necessary at all. Haskell is a big language, and some of those other sources will begin with things that will not be relevant for us.

If and when you really want one, a good textbook for many purposes is *Programming in Haskell* by Graham Hutton (Cambridge University Press). The best way to use this would be as an occasional reference throughout the second half of the course, rather than as an initial learning aid.

Course Meetings and Assignments

Lectures and Discussion Sections

All lectures, sections, and office hours will be held in-person unless the university dictates otherwise. In order to facilitate open and active discussion, course meetings will not generally be recorded. Exceptions may be made for students who require recordings as part of accommodations that have been registered with the Center for Accessible Education (CAE). If you need to miss class for another reason, you are encouraged to get notes from a classmate.

You will not be graded on your attendance in lecture or discussion, but **regular attendance is crucial to your mastery of the course material and to your success in this class**. Lectures will present the core concepts each week with some illustrative examples, and discussion sections will provide additional examples, coding practice, and review. This course moves very quickly, particularly as the quarter progresses. If you do not attend lectures and sections, you will find yourself falling behind.

Participating in class is highly encouraged and valuable for everyone. It makes you an active learner and greatly increases your understanding of the material, it benefits your classmates who may have had the same questions, and it benefits the instructors in tailoring the course material to better serve your learning needs. Active participation means speaking up when you find something unclear (or think there might be a bug or typo!), when you see a connection between a particular topic and other material we have discussed in this course, and when you are unconvinced by the logic of an argument or have a different argument that you would like to propose. All discussion is expected to be respectful and inclusive towards the different identities, backgrounds, and perspectives of your fellow students and instructors.

Homework Assignments (about 8; 70%)

There will be roughly one homework assignment per week, due at the end of the week (11:59pm on Fridays). These will provide practice working through the core ideas that we are covering each week, coding some small programs, and extending these core ideas to related topics. In order to build in some flexibility, your lowest homework score out of the **first three** homework scores will be dropped from your final grade. The remaining homework scores will be equally weighted together to comprise 70% of the total grade for the class.

The purpose of homework assignments is to advance and solidify your own understanding of the course content. Partial responses in which you clearly identify what is missing will in general receive more credit than responses that try to cover up these gaps.

Final Project (30%)

The final project for this course will be due at 11:59pm on Friday, December 13th. This will involve two components: (1) investigating the formal properties of particular grammatical systems, either in code or by hand, and (2) writing a short paper (about 8-12 pages double-spaced) that explains your work and connects it to bigger-picture ideas in the course. More details will come around the middle of the quarter.

Assignment Submission Policies

All written assignments must be typed and submitted as a PDF via BruinLearn (not email) in order to receive credit. For assignments with a coding component, **the code must compile** in order to receive credit. Please comment out any partial solutions that don't compile.

Any assignment turned in more than 10 minutes after the submission deadline will be considered late. We will accept late homework assignments (not the final project) up to 48 hours after the deadline. Assignments submitted up to 24 hours late will receive a grade deduction of 1 point (5% of the 20 possible points). Assignments submitted between 24 and 48 hours late will receive a grade deduction of 2 points (10% of the 20 possible points). After 48 hours after the deadline, late homework will not be accepted and will be assigned a zero. The final project must be submitted by the deadline in order to receive credit.

Exceptions to this policy may be made for extenuating circumstances that are brought to my attention in a timely manner (e.g., illness, COVID-related disruptions, jury duty, bereavement, or religious observances). If you are experiencing circumstances that are affecting your ability to keep up with assignments, please reach out to me as soon as possible, and I will be more than happy to help. Please try your utmost to contact me *before* the relevant assignment deadline has passed.

You are permitted, and indeed encouraged, to discuss assignments with your classmates. However, all assignments that you submit must be written by you individually, in your own words, and reflect your own understanding of the course material. We will not permit direct copies or jointly authored assignments. All discussions should abide by both the letter and the spirit of the “whiteboard policy.” You may work together on a “whiteboard” and discuss things as long as you wish and in as much detail as you wish, but then you must “erase the whiteboard” and not take any written notes away from the discussion. The idea is that being able to write up your own solution individually establishes that you understand what you have submitted.

Policy on Generative AI

You are permitted to use generative AI (such as ChatGPT and similar tools) to ask questions about material from lecture or sections, or about examples of code that we have worked through in class. However, this is not recommended and is not generally a useful way to get helpful guidance: your instructor, TA and classmates are better resources. **You are not permitted to use an AI to solve homework problems or write assignments.** Be very careful about the questions that you ask. If you ask for hints or explanations of homework problems or final project prompts, the AI will often give you “hints” or “explanations” that resemble solutions or suggested responses. Making use of those suggestions is not permitted, because this means that you are not writing your assignment individually, in a way that reflects your own understanding of the material. If you consult an AI at any point in completing an assignment, then you are required to submit a transcript of the complete set of questions that you asked and the complete set of responses that it provided, in order to demonstrate that you used it in a permitted way. Any unauthorized use of AI will be treated as academic dishonesty.

Other Procedures and Policies

How to Contact Us

Please make use of lectures, sections, and office hours for questions about course material and assignments. We will also set up an online forum on Campuswire to facilitate discussions. You should not reveal answers to assignment questions on the forum—please don't spoil the fun for others!

For logistical or administrative matters, please **email us with "185A" somewhere in the subject line**. If you are in doubt about whether to email the instructor or TA for a particular question, include us both. Note that emails received on weekdays after 5pm or on weekends will probably not receive a prompt response. Please plan ahead so as to minimize your need to reach us quickly after hours.

How We'll Contact You

We will send important announcements via the email list on BruinLearn. If you are registered for the class, you should already be on the email list. Keep in mind that all emails will be sent to the email account registered with UCLA, and be sure to check this account regularly.

COVID-19

For the safety of the whole class, all students must consistently follow the current university directives related to COVID-19 mitigation, which can be found at <https://covid-19.ucla.edu/ucla-return-to-campus/>. If you need to miss class to isolate or quarantine, or for any other extenuating circumstance, please get in touch with me as soon as possible and we'll work together to find a way for you to keep up with course material. In addition, course policies and requirements may need to change during the quarter to respond to any developments in COVID-19. I appreciate your patience and flexibility.

Academic Integrity

All members of the UCLA academic community are expected to maintain standards of academic honesty, specified by the [UCLA Student Conduct Code](#). It is important for you to know what constitutes improper academic conduct and the consequences for this misconduct. What academic integrity means in this course: students are permitted, and indeed encouraged, to discuss course material and assignments together. However, all work you submit in this course must be written by you individually. Any sources that you consult must be properly cited. Any idea, code, or information that is not cited is assumed to be your individual contribution. **Using code or ideas from any source and presenting it as your own work is plagiarism.** This includes using online sources or AI (such as ChatGPT) to write your assignments. Any incidents of suspected academic dishonesty, including copying work or allowing someone to copy your work, will be referred to the Dean of Students.

Use of Electronic Devices in Class

We will frequently be using computers in class. However, electronic devices can be disruptive to your learning and that of your classmates if used inappropriately. Students are expected to limit their use of such devices to classroom activities only, such as taking notes and participating in coding examples. Any other activities (checking email, texting, social media, etc.) are inappropriate during class time.

Diversity and Self-Identification

It is my intent that students from all backgrounds and perspectives be well served by this course and that the diversity that students bring to this class be viewed as a resource and strength. I am committed to fostering a learning environment that honors your identities, including race, ethnicity, gender, sexuality, religion, national origin, age, and dis/ability. Presented materials and discussion are meant to be respectful of diversity and different identities. Your suggestions are encouraged and appreciated. Please let me know of ways to improve the effectiveness of the course for you personally or for other students or groups.

I invite you to tell us how you want to be referred to both in terms of your names and your pronouns. All aspects of your identity are your choice to disclose and should be self-identified, not presumed or imposed. I will do my best to address and refer to all students accordingly, and I ask you to do the same for your fellow colleagues.

Students with Disabilities

Any students seeking disability services should register with the Center for Accessible Education (CAE) at caeintake@saonet.ucla.edu or (310) 825-1501 as soon as possible. When possible, students should contact the CAE within the first two weeks of the term, as reasonable notice is needed to coordinate accommodations. If you are already registered and require assistance for the quarter, please release your accommodation letters by the end of Week One. For more information, visit www.cae.ucla.edu.

Title IX

Title IX prohibits gender discrimination, including sexual harassment, domestic and dating violence, sexual assault, and stalking. If you have experienced any of these forms of discrimination, harassment, or violence, you can receive confidential support and advocacy at the [CARE Advocacy Office for Sexual and Gender-Based Violence: CAREadvocate@careprogram.ucla.edu](mailto:CAREadvocate@careprogram.ucla.edu), (310) 206-2465. In addition, [Counseling and Psychological Services \(CAPS\)](#) provides confidential counseling to all students and can be reached 24/7 at (310) 825-0768.

You can also non-confidentially report sexual violence or sexual harassment directly to the University's Title IX Coordinator at titleix@conet.ucla.edu, (310) 206-3417. Reports to law enforcement can be made to UCPD at (310) 825-1491. Faculty and TAs are required under the UC Policy on Sexual Violence and Sexual Harassment to inform the Title IX Coordinator should they become aware that you or any other student has experienced sexual violence or sexual harassment.

Mental Health and Student Support

University can be difficult, and many students find themselves struggling at some time or another. But you don't need to struggle alone: there are many people on campus whose job it is to help students with all kinds of circumstances. Some of these resources include [Counseling and Psychological Services \(CAPS\)](#), the [Ashe Student Health and Wellness Center](#), the [Consultation and Response Team](#), the office of [Equity, Diversity and Inclusion](#). If you are experiencing challenges that are having an impact on your participation in this course, please talk to me and we can work together to create a plan. If you're not comfortable talking to me directly, please reach out to an advisor or to other support services on campus.